



Author: HAF

#### Note from author

These notes are mostly based on the syllabus and past papers

Fair warning: they might look short (2–3 pages), but I've condensed everything down to keep the important stuff while removing the extra yap. That said, I wouldn't recommend using these as quick revision notes, but think of them as detailed notes covering almost everything

Spotted a mistake? Suggestions?

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# Introduction

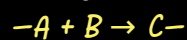
Types of reactions:

## 1. Addition

What happens: Two reactants combine to form a single product. No atoms are lost.

Key clue: The reactants are usually unsaturated (have a double/triple bond).

Example: Alkenes reacting with hydrogen ( $\text{H}_2$ ), bromine ( $\text{Br}_2$ ), or steam ( $\text{H}_2\text{O}$ ).



## 2. Elimination

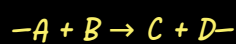
What happens: A small molecule (like  $\text{H}_2\text{O}$  or  $\text{HCl}$ ) is removed (eliminated) from a larger molecule, often forming a double bond.

Example: Heating a halogenoalkane or alcohol with  $\text{NaOH}/\text{OH}^-$  (or conc.  $\text{H}_2\text{SO}_4/\text{H}_3\text{PO}_4$  for alcohols) to form an alkene.

## 3. Substitution

What happens: An atom (or group) is swapped for a different atom (or group).

Example: Halogenoalkane +  $\text{NaOH}$  (aqueous)  $\rightarrow$  Alcohol (Halogen swapped for  $\text{OH}$ ).



## 4. Oxidation

What happens: Gain of oxygen, loss of hydrogen, or loss of electrons. (Usually increases the oxygen/halogen content).

Example: Primary alcohol  $\rightarrow$  Aldehyde  $\rightarrow$  Carboxylic acid.

## 5. Reduction (a type of addition reaction)

What happens: Gain of hydrogen, loss of oxygen, or gain of electrons. (Opposite of oxidation).

Example: Alkenes to alkanes (using  $\text{H}_2/\text{Ni}$  catalyst) or Nitriles to amines (using  $\text{LiAlH}_4$  or  $\text{H}_2/\text{Ni}$ ).

## 6. Hydrolysis

What happens: A reaction with water (or aqueous  $\text{OH}^-$ ) that breaks a chemical bond.

Example: Halogenoalkanes + aqueous  $\text{NaOH}$  (or water/ $\text{AgNO}_3$ )  $\rightarrow$  Alcohols.

## 7. Polymerisation

What happens: Many small molecules (monomers) join together to form a giant chain (polymer).

Example: Addition polymerisation of alkenes (e.g. ethene  $\rightarrow$  poly(ethene)).

## Free Radical Substitution

Definition

A reaction mechanism involving free radicals (species with unpaired electrons) where bonds are broken by homolytic fission.

General Equation

Definition: Free Radical

A species with an unpaired electron.

Bond Breaking: Homolytic Fission

Each atom takes one electron from the covalent bond.

Shown using half-headed curly arrows (fishhooks).

Products are free radicals.

## Electrophilic substitution

Definition

A reaction mechanism in which an electrophile attacks a double bond, causing it to open and add two atoms/groups to the carbons.

General Equation

Definition: Electrophile

An electron pair acceptor. An ion or molecule that accepts a pair of electrons to form a new covalent bond.

Bond Breaking: Heterolytic Fission

Both electrons go to one atom.

Products are ions (positive and negative species).

Leads to electrophiles and nucleophiles.

## Nucleophilic Substitution

Definition

A reaction mechanism in which a nucleophile donates a lone pair to a electron-deficient carbon, swapping the halogen for another atom/group.

General Equation

Definition: Nucleophile

An electron pair donor. An ion or molecule that donates a lone pair of electrons to form a new covalent bond (nucleus loving)

Bond Breaking: Heterolytic Fission

Both electrons go to one atom (the halogen).

Products are ions.

link between Bond Polarity and Mechanism:

Polar bonds (e.g.,  $\text{C-X}$  in halogenoalkanes) undergo heterolytic fission. The electrons are pulled towards the more electronegative atom. This creates ions ( $\delta^+$  and  $\delta^-$ ) and leads to ionic mechanisms (electrophilic/nucleophilic).

Non-polar bonds (e.g.,  $\text{Cl-Cl}$  in alkanes) can undergo homolytic fission (usually under UV light). Electrons are shared equally, so each atom takes one. This creates free radicals and leads to radical mechanisms.

# Halogenoalkanes

## Intro to halogenoalkanes

Halogenoalkanes are alkanes where one or more hydrogen atoms have been replaced by a halogen atom (F, Cl, Br, I).

General formula:  $C_nH_{2n+1}X$

## Nomenclature

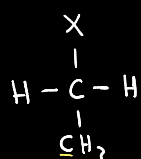
The naming system has already been explained in previous notes (alkanes U1) but here are some examples:

| Name            | Formula          | How to name it           |
|-----------------|------------------|--------------------------|
| Chloromethane   | $CH_3Cl$         | Chloro + methane         |
| Bromoethane     | $C_2H_5Br$       | Bromo + ethane           |
| 1-Chloropropane | $CH_3CH_2CH_2Cl$ | Chloro on C-1 of propane |
| 2-Bromopropane  | $CH_3CHBrCH_3$   | Bromo on C-2 of propane  |

## Classification

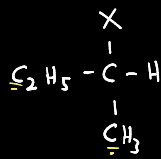
Halogenoalkanes can be classified as primary, secondary or tertiary depending on the number of carbon atoms attached to the C-X functional group

Primary ( $1^\circ$ )



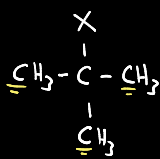
The carbon attached to the halogen is bonded to 1 carbon atom and 2 hydrogens.

Secondary ( $2^\circ$ )



The carbon attached to the halogen is bonded to 2 carbon atoms and 1 hydrogen.

Tertiary ( $3^\circ$ )



The carbon attached to the halogen is bonded to 3 carbon atoms and 0 hydrogens.

## Reactions

Unlike alkane halogenoalkanes are reactive molecules because they have **Polar Bond: The C-X bond is polar** (C are  $\delta^+$  while X is  $\delta^-$ ). The carbon is electron-deficient and wants to be attacked by electron-rich species.

Depending on the conditions (reagent and solvent), halogenoalkanes can react in two different ways:

| Pathway                   | What happens                                                            | Conditions                                                                        |
|---------------------------|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Nucleophilic Substitution | The halogen is swapped for another atom/group.                          | Aqueous conditions ( $OH^-$ in water) or with nucleophiles like $CN^-$ , $NH_3$ . |
| Elimination               | The halogen and a hydrogen are removed, forming a double bond (alkene). | Ethanol conditions ( $OH^-$ in ethanol) with heat.                                |

## The Role of the Hydroxide Ion

The hydroxide ion ( $OH^-$ ) is special because it can act in two different ways:

- As a Nucleophile: Donates a lone pair to form a new bond. Happens in aqueous conditions. Product = Alcohol
- As a Base: Accepts a proton ( $H^+$ ) from the carbon next to the halogen. Happens in ethanolic conditions. Product = Alkene

Key Point: The solvent (water vs ethanol) decides the role of  $OH^-$ :

Aqueous: Substitution

Alcoholic: Elimination

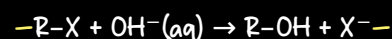
(more on that later)

Recall:

Nucleophiles are electron pair donors (nucleus loving)

## Reaction with Aqueous Potassium Hydroxide (KOH)

- Reagent: Potassium hydroxide dissolved in water (aqueous).
- Conditions: Heat under reflux.
- Product: Alcohol
- Role of  $OH^-$  (reagent): Nucleophile [Mechanism type: Nucleophilic substitution]



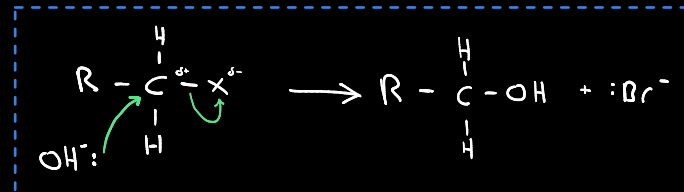
The carbon attached to the halogen is  $\delta^+$  (due to electronegative halogen). The hydroxide ion ( $OH^-$ ) is electron-rich (negative charge and lone pairs). Opposites attract.

Mechanism: Primary halogenoalkane reacts with aqueous alkali

Bond Forms: lone pair on  $OH^-$  is donated to the  $\delta^+$  carbon forming new covalent bond

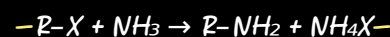
Bond Breaks: As the new bond forms, the C-X bond breaks heterolytically and the electron pair leave with the bromine to form  $X^-$

Products: Alcohol + anion  $X^-$



## Reaction with Alcoholic Ammonia ( $NH_3$ )

- Reagent: Ammonia dissolved in ethanol (alcoholic ammonia).
- Conditions: Heat excess ammonia in a sealed tube under pressure in an alcoholic solution.
- Product: Amine (specifically a primary amine)
- Role of  $NH_3$  (reagent): Nucleophile [Mechanism type: Nucleophilic substitution]



Why is a sealed tube used instead of reflux?

-If you try to heat a mixture of halogenoalkane and ammonia under reflux, the ammonia will boil away as a gas and escape through the condenser, therefore, a sealed tube is used since as you heat the tube, the ammonia vaporises but cannot escape.

Further Substitution: Why excess ammonia is essential?

- The Reactant: Ammonia ( $:NH_3$ ) is a nucleophile. It attacks the halogenoalkane to form a primary amine ( $R-NH_2$ ).
- The Product: The primary amine ( $R-NH_2$ ) still has a lone pair on the nitrogen.
- The Consequence: The amine product can now act as a nucleophile and attack another halogenoalkane molecule (low yield).

| Step | Reaction                   | Product                                  |
|------|----------------------------|------------------------------------------|
| 1    | $R-X + NH_3 \rightarrow$   | Primary amine ( $R-NH_2$ ) + $HX$        |
| 2    | $R-X + R-NH_2 \rightarrow$ | Secondary amine ( $R_2NH$ ) + $HX$       |
| 3    | $R-X + R_2NH \rightarrow$  | Tertiary amine ( $R_3N$ ) + $HX$         |
| 4    | $R-X + R_3N \rightarrow$   | Quaternary ammonium salt ( $R_4N^+X^-$ ) |

Mechanism: Primary halogenoalkane reacts with ammonia

### Step 1: Nucleophilic Attack

Bond Forms: lone pair on  $NH_3$  is donated to the  $\delta^+$  carbon forming new covalent bond

Bond Breaks: As the new bond forms, the C-X bond breaks heterolytically and the electron pair leave with the bromine to form  $X^-$

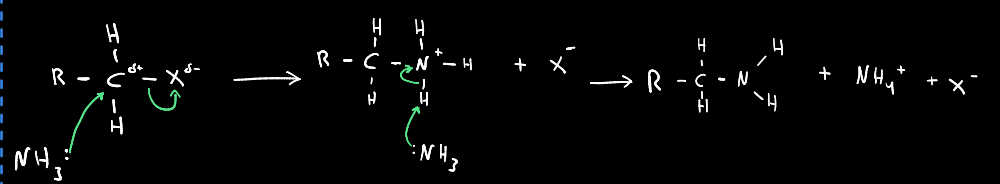
Intermediate: alkylammonium ion

### Step 2: Proton Transfer

Bond Forms: Lone pair on a second ammonia molecule ( $:NH_3$ ) donates to a proton ( $H^+$ ) on the alkylammonium ion.

Bond Breaks: The N-H bond in the alkylammonium ion breaks heterolytically, with the electron pair staying on the nitrogen.

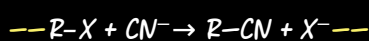
Products: Neutral primary amine ( $CH_3CH_2NH_2$ ) + Ammonium ion ( $NH_4^+$ )



# Halogenoalkanes

## Reaction with Alcoholic Potassium Cyanide (KCN)

- Reagent: Potassium cyanide dissolved in ethanol (alcoholic)
- Conditions: Heat under reflux.
- Product: Nitrile
- Role of  $\text{CN}^-$  (reagent): Nucleophile [Mechanism type: Nucleophilic substitution]



Example:

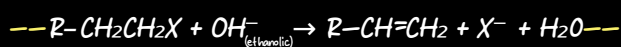


Notice that a carbon was added to the chain; the nitrile ( $-\text{CN}$ ) introduces an extra carbon atom. This is useful for increasing carbon chain length in organic synthesis.

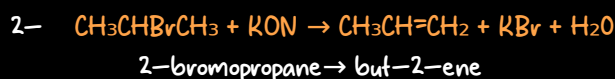
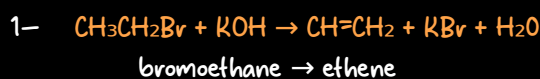
The carbon attached to the halogen is  $\delta^+$  (due to electronegative halogen). The cyanide ion ( $\text{CN}^-$ ) is electron-rich (negative charge and lone pairs). Opposites attract.

## Reaction with Ethanolic Potassium Hydroxide

- Reagent: Potassium hydroxide dissolved in ethanol
- Conditions: Heat under reflux.
- Product: Alkene
- Role of  $\text{OH}^-$  (reagent): Base (accepts a proton, does not attack the carbon) [Mechanism type: Elimination]



Example:



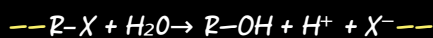
In this reaction, the  $\text{OH}^-$  acts as a base accepting a proton ( $\text{H}^+$ ) from the carbon adjacent to the one bearing the halogen. The  $\text{C-X}$  bond breaks heterolytically forming  $\text{X}^-$ , and a  $\text{C=C}$  double bond forms.

**Note:** If the halogenoalkane is unsymmetrical (the two carbons adjacent to the  $\text{C-OH}$  are different), there are two possible hydrogens that can be removed. This leads to two different alkene products.

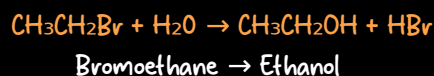
The solvent matters!  
Aqueous  $\text{KOH} \rightarrow$  substitution (alcohol);  
Ethanolic  $\text{KOH} \rightarrow$  elimination (alkene)

## Reaction with water (Hydrolysis)

- Reagent: Water ( $\text{H}_2\text{O}$ )
- Product: Alcohol
- Role of  $\text{H}_2\text{O}$  (reagent): Nucleophile [Mechanism type: Nucleophilic substitution]



Example:



This reaction is slow because water is a weak nucleophile (unlike  $\text{OH}^-$ ).

## Compare Rates of Hydrolysis

Using Aqueous Silver Nitrate in Ethanol

What is the reagent?

Water ( $\text{H}_2\text{O}$ ): The **actual nucleophile** that reacts with the halogenoalkane.

Ethanol ( $\text{C}_2\text{H}_5\text{OH}$ ): The **co-solvent** that helps dissolve the **mixture and makes it homogeneous**. Halogenoalkanes are non-polar and would not mix well with water alone).

Silver nitrate ( $\text{AgNO}_3$ ): The hydrolysis reaction releases halide ions ( $\text{X}^-$ ), but we cannot see them. Adding silver nitrate ( $\text{AgNO}_3$ ) causes a silver halide **precipitate** ( $\text{AgX}$ ) to form. The **time taken for the precipitate to appear indicates the rate of hydrolysis** — faster precipitate = faster hydrolysis.

## Comparison (i): Primary, Secondary and Tertiary halogenoalkanes

**Aim:** To compare the rate of hydrolysis of halogenoalkanes with the same halogen but different structures.

**Method:** Add aqueous silver nitrate in ethanol to each isomer at room temperature. Observe the time taken for a precipitate to form.

| Type      | Example                                      | Relative Rate | Observation                                              |
|-----------|----------------------------------------------|---------------|----------------------------------------------------------|
| Tertiary  | $(\text{CH}_3)_3\text{CBr}$                  | Fastest       | Precipitate forms <b>immediately</b> at room temperature |
| Secondary | $\text{CH}_3\text{CHBrCH}_3$                 | Medium        | Precipitate forms within <b>minutes</b>                  |
| Primary   | $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$ | Slowest       | Precipitate forms <b>slowly</b> ; may require warming    |

Order of Reactivity: Tertiary > Secondary > Primary

## Comparison (ii): Chloro-, Bromo- and Iodoalkanes

**Aim:** To compare the rate of hydrolysis of halogenoalkanes with different halogens.

**Method:** Add aqueous silver nitrate in ethanol to each isomer at room temperature. Observe the time taken for a precipitate to form.

| Halogenoalkane | Bond          | Bond Strength (kJ/mol) | Relative Rate | Observation                                        |
|----------------|---------------|------------------------|---------------|----------------------------------------------------|
| Iodoalkane     | $\text{C-I}$  | 238                    | Fastest       | Precipitate forms quickly at room temperature      |
| Bromoalkane    | $\text{C-Br}$ | 276                    | Medium        | Precipitate forms slower than iodo                 |
| Chloroalkane   | $\text{C-Cl}$ | 338                    | Slowest       | Precipitate forms very slowly; may require warming |

Order of Reactivity: Iodo > Bromo > Chloro

Explanation:

As you go down Group 7, **atomic size increases**  $\rightarrow$  **bond length increases**  $\rightarrow$  **bond strength decreases**.

The **weaker the  $\text{C-X}$  bond (lower bond enthalpy)**  $\rightarrow$  the **easier the halogen leaves**, so the faster the hydrolysis.

$\text{C-I}$  is the weakest bond  $\rightarrow$  fastest reaction.

$\text{C-Cl}$  is the strongest bond  $\rightarrow$  slowest reaction.

**Bond Polarity  $\neq$  Reactivity**

## Fair testing

When comparing rates of hydrolysis, **all other factors must be kept constant** to ensure a fair comparison:

- **Concentration:** **Same concentration of halogenoalkane and silver nitrate** ensures rate differences are due to structure or halogen, not concentration.
- **Temperature:** All reactions should be at the **same temperature** (e.g. room temperature) because higher temperature increases rate regardless of the halogenoalkane.
- **Solvent composition:** Same ratio of water to ethanol ensures the solvent environment is identical for each reaction.
- **Volume of solution:** Same total volume ensures consistent mixing and observation.
- **Same halogen type** [For comparison (i)]: e.g. all are bromoalkanes
- **Same structure** [for comparison (ii)]: e.g. all are primary halogenoalkanes

Why this matters: By keeping these factors constant, any difference in the time taken for precipitate to form can be attributed only to:

1) The structure of the halogenoalkane (primary/secondary/tertiary)

**OR**

2) The halogen present ( $\text{Cl/Br/I}$ )



# Alcohols

## Intro to alcohols

Alcohols are organic compounds containing a hydroxyl group ( $\text{-OH}$ ) bonded to a saturated carbon atom.

General Formula:  $\text{C}_n\text{H}_{2n+1}\text{OH}$  (or  $\text{C}_n\text{H}_{2n+2}\text{O}$ )

## Nomenclature

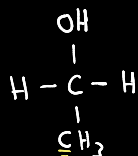
The naming system has already been explained in previous notes (alkanes U1) but here are some examples:

| Name        | Formula                                      | How to name it               |
|-------------|----------------------------------------------|------------------------------|
| Methanol    | $\text{CH}_3\text{OH}$                       | methan + ol                  |
| Ethanol     | $\text{C}_2\text{H}_5\text{OH}$              | ethan + ol                   |
| Propan-1-ol | $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ | propan + ol (OH on carbon 1) |
| Propan-2-ol | $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$ | propan + ol (OH on carbon 2) |

## Classification

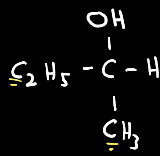
Alcohols can be classified as primary, secondary or tertiary depending on the number of carbon atoms attached to the carbon bearing  $\text{-OH}$  group.

Primary ( $1^\circ$ )



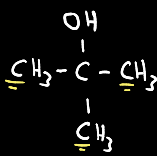
The carbon attached to the  $\text{-OH}$  is bonded to 1 carbon atom and 2 hydrogens.

Secondary ( $2^\circ$ )



The carbon attached to the  $\text{-OH}$  is bonded to 2 carbon atoms and 1 hydrogen.

Tertiary ( $3^\circ$ )



The carbon attached to the  $\text{-OH}$  is bonded to 3 carbon atoms and 0 hydrogens.

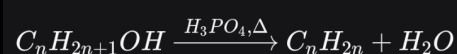
## Dehydration (Elimination) with Concentrated Phosphoric Acid

Alcohols undergo elimination (dehydration) to form alkenes. A water molecule is removed from the alcohol, creating a  $\text{C}=\text{C}$  double bond.

**Reagent:** Concentrated phosphoric acid ( $\text{H}_3\text{PO}_4$ ) — acts as a catalyst and dehydrating agent

**Conditions:** Heat

General Equation:



During dehydration, the  $\text{-OH}$  group and a hydrogen atom from an adjacent carbon are removed as water. A  $\text{C}=\text{C}$  double bond forms between the two carbons.

If the alcohol is unsymmetrical (the two carbons adjacent to the  $\text{C-OH}$  are different), there are two possible hydrogens that can be removed. This leads to two different alkene products.

Alternative:  $\text{Al}_2\text{O}_3$  can also be used as a dehydrating agent. The alcohol is passed as a vapour over hot  $\text{Al}_2\text{O}_3$  ( $300\text{--}350^\circ\text{C}$ ) to form the alkene.

## Halogenating Agents

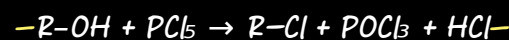
Alcohols can be converted into halogenoalkanes by replacing the  $\text{-OH}$  group with a halogen atom (Cl, Br, or I).

### Producing Chloroalkanes

**Reagent:** Phosphorus(V) chloride ( $\text{PCl}_5$ )

**Product:** Chloroalkane

General Equation:



**Observations:**

- Steamy/misty white fumes of  $\text{HCl}$  are produced
- This is used as a qualitative test for the presence of the  $\text{-OH}$  group

Chlorination of tertiary alcohols can be done in a different method (both methods work), that does not work with primary and secondary alcohols, and that one is by mixing (shaking) the alcohol with hydrochloric acid at room temperature.

A common example is:  $(\text{CH}_3)_3\text{COH} + \text{HCl} \rightarrow (\text{CH}_3)_3\text{CCl} + \text{H}_2\text{O}$

### Producing Bromoalkanes

**Reagent:** Potassium bromide ( $\text{KBr}$ ) + 50% concentrated sulfuric acid ( $\text{H}_2\text{SO}_4$ )

**Product:** Bromoalkane

**What happens:** The  $\text{H}_2\text{SO}_4$  reacts with  $\text{KBr}$  to produce  $\text{HBr}$  *in situ* (within the reaction mixture) then  $\text{HBr}$  reacts with the alcohol to form the bromoalkane.

**Important:** 50% concentrated sulfuric acid is used as more concentrated sulfuric acid would oxidise bromide ions to bromine ( $\text{Br}_2$ ), resulting in different products (e.g. bromine instead of  $\text{HBr}$ , leading to possible side reactions).

**Step 1: Formation of  $\text{HBr}$ :**



**Step 2: Substitution with Alcohol:**



*In situ* (Latin: "in place"):

A reagent is produced within the reaction mixture and reacts immediately, rather than being added as a separate reagent.

## Alcohols: Reactions

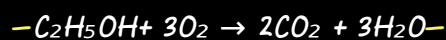
### Combustion (Reaction with Oxygen in Air)

Alcohols undergo complete combustion in excess oxygen to produce carbon dioxide and water. This reaction is **exothermic** (releases heat).

General Equation:



Example (Ethanol):



**Observations:**

- Burns with a clean, blue flame (similar to alkanes)
- Produces heat energy — used as a fuel (e.g., ethanol in spirit burners, biofuel)

# Alcohols

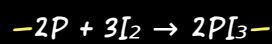
## Producing Iodoalkanes

**Reagent:** Red phosphorus and iodine ( $I_2$ )

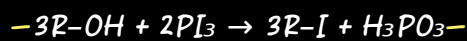
**Product:** Iodoalkane

**What happens:** Red phosphorus and iodine react together to form phosphorus(III) iodide ( $PI_3$ ) *in situ* (within the reaction mixture), which then reacts with the alcohol.

**Step 1: Formation of  $PI_3$ :**



**Step 2: Substitution with Alcohol:**



## Oxidation

**Reagent:** acidified potassium dichromate(VI) ( $K_2Cr_2O_7$  + dilute  $H_2SO_4$ )

**Observation** (if alcohol is oxidized):

—Solution color changes from **orange** [dichromate(VI) ion ( $Cr_2O_7^{2-}$ )] to **green** [Chromium(III) ions ( $Cr^{3+}$ )]

**What Happens During Oxidation?**

Oxidation of alcohols involves the **removal of hydrogen atoms** from the molecule.

Specifically:

- The  $-OH$  group loses its hydrogen atom.
- The carbon atom bearing (not carbon the adjacent) the  $-OH$  group loses one hydrogen atom (if available)

These two hydrogens combine with oxygen from the oxidising agent to form water ( $H_2O$ ).

General reaction:  $Alcohol \xrightarrow{[O]} Aldehyde/Ketone + H_2O$

For oxidation to occur, the carbon bearing the  $-OH$  group **must have at least one hydrogen atom** attached to it.

| Alcohol Type            | Structure  | Hydrogens on C-OH | Oxidation Possible?         |
|-------------------------|------------|-------------------|-----------------------------|
| Primary ( $1^\circ$ )   | $R-CH_2OH$ | 2 hydrogens       | Yes — can be oxidised twice |
| Secondary ( $2^\circ$ ) | $R_2CHOH$  | 1 hydrogen        | Yes — can be oxidised once  |
| Tertiary ( $3^\circ$ )  | $R_3COH$   | 0 hydrogens       | No — no hydrogen to remove  |

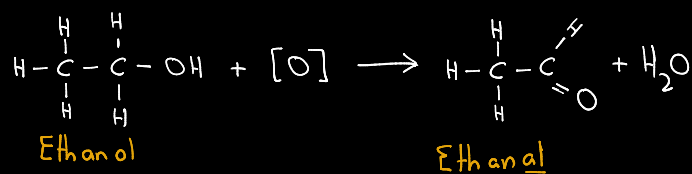
## Primary Alcohols: Partial Oxidation

A primary alcohol **loses two hydrogen atoms** (one from the  $-OH$  group and one from the carbon) to form an aldehyde.

These aldehydes can rapidly **get oxidized further** to carboxylic acids → that's why to obtain aldehydes we use **distillation with addition** (explained in more details in later sections)

**General equation:**  $R-CH_2OH + [O] \rightarrow R-CHO + H_2O$

**Example:**



## Primary Alcohols: Full Oxidation

A primary alcohol is **oxidised twice** — first to an aldehyde, then further to a carboxylic acid.

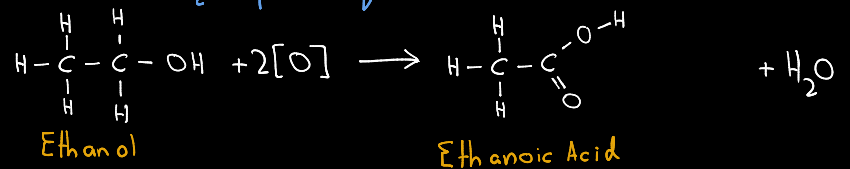
To achieve this we carry out heating under reflux (explained in more details in later sections)

**General equation:**  $R-CH_2OH + [O] \rightarrow R-CHO + H_2O$

**then**  $R-CHO + [O] \rightarrow R-COOH$

**Example:**

[Simplified equation]

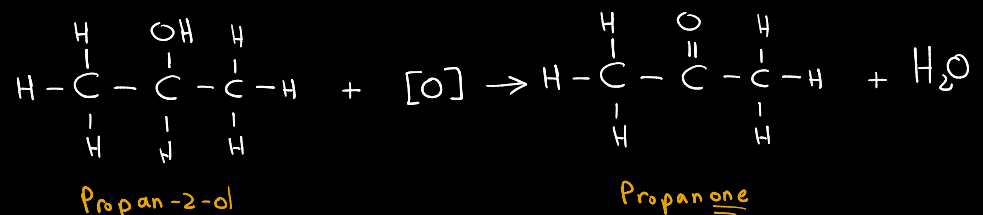


## Secondary Alcohols

A secondary alcohol **loses two hydrogen atoms** (one from the  $-OH$  group and one from the carbon) to form a ketone. Unlike primary alcohols, ketones **cannot be oxidised further**.

**General equation:**  $R_2CHOH + [O] \rightarrow R_2C=O + H_2O$

**Example:**



## Tests

| Step | Action                                                      | Observation                            | Conclusion                        | Chemical Explanation                                     |
|------|-------------------------------------------------------------|----------------------------------------|-----------------------------------|----------------------------------------------------------|
| 1    | Add acidified $K_2Cr_2O_7$                                  | Orange remains                         | Tertiary alcohol                  | No H on C-OH → cannot be oxidised                        |
|      |                                                             | Orange → Green                         | Primary or secondary              | Oxidation occurred → $Cr_2O_7^{2-}$ reduced to $Cr^{3+}$ |
| 2    | <b>Choose technique based on what you want to find out:</b> |                                        |                                   |                                                          |
|      | <b>Distillation with addition</b> (collect product)         | Product collected                      | —                                 | Aldehyde or ketone is distilled off as formed            |
|      | <b>Heat under reflux</b> (full oxidation)                   | Product in flask                       | —                                 | Aldehyde (if formed) remains and is further oxidised     |
| 3a   | Add Benedict's/Fehling's to distilled product (heat gently) | <b>Brick-red precipitate</b>           | Aldehyde → Primary alcohol        | Aldehyde reduces $Cu^{2+}$ to $Cu_2O$ (brick-red)        |
|      |                                                             | No change (blue remains)               | Ketone → Secondary alcohol        | Ketone cannot reduce $Cu^{2+}$                           |
| 3b   | Add $Na_2CO_3$ or $NaHCO_3$ to reflux mixture               | <b>Effervescence</b> ( $CO_2$ bubbles) | Carboxylic acid → Primary alcohol | Acid + carbonate → $CO_2$ gas                            |
|      |                                                             | No change                              | Ketone → Secondary alcohol        |                                                          |

## Other tests

### Tollens reagent

**Reagent:** Aqueous ammonia with silver nitrate

**Detects:** Aldehydes

**Conditions:** Warm gently

**Observation:** Silver mirror forms on test tube walls

**Explanation:** Aldehyde is oxidised to carboxylic acid;  $Ag^+$  is reduced to  $Ag(s)$

**Ketones:** No reaction

### Reaction with sodium

**Reagent:** Sodium metal Na

**Detects:**  $-OH$  group (alcohols, carboxylic acids...)

**Conditions:** Room temperature, dry conditions

**Observation:** Effervescence (bubbles of  $H_2$  gas) + sodium disappears

**Confirmation:** Squeaky pop test with a lighted splint

**Equation:**  $2R-OH + 2Na \rightarrow 2R-ONa + H_2$